

ICAI-SR: Item Categorical Attribute Integrated Sequential Recommendation

Xu Yuan, Dongsheng Duan*
yuanxu19g@ict.ac.cn, dds@cert.org.cn
ICT-CAS, CNCERT
Beijing, China

Lingling Tong, Lei Shi
tongling300@sina.com, sl@cert.org.cn
CNCERT
Beijing, China

Cheng Zhang
zhangcheng@ict.ac.cn
ICT-CAS
Beijing, China

ABSTRACT

Sequential recommendation (SR) has attracted much research attention in the past few years. Most existing attribute integrated SR models do not directly model the complex relations between items and categorical attributes, as well do not exploit the power of attribute sequence in predicting the next item. In this paper, we propose an **Item Categorical Attribute Integrated Sequential Recommendation (ICAI-SR)** framework, which consists of an Item-Attribute Aggregation (IAA) model and Entity Sequential (ES) models. In IAA model, we employ a heterogeneous graph to represent the complex relations between items and different types of categorical attributes, then the attention mechanism based neighborhood aggregation is designed to model the correlations between items and attributes. For ES models, there are one Item Sequential (IS) model and one or more Attribute Sequential (AS) models. With IS and AS models, not only the item sequence but also the attribute sequence are used to predict the next item during model training. ICAI-SR is instantiated by taking Gated Recurrent Unit (GRU) and Bidirectional Encoder Representations from Transformers (BERT) as ES models, resulting in ICAI-GRU and ICAI-BERT respectively. Extensive experiments have been conducted on three public datasets to validate the performance of ICAI-SR. Experimental Results show that ICAI-SR performs better than both basic SR models and a competitive attribute integrated SR model.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning approaches.**

KEYWORDS

Sequential Recommendation; Recurrent Neural Network; Heterogeneous Graph; Graph Neural Network; Attention Mechanism

ACM Reference Format:

Xu Yuan, Dongsheng Duan, Lingling Tong, Lei Shi, and Cheng Zhang. 2021. ICAI-SR: Item Categorical Attribute Integrated Sequential Recommendation. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '21)*, July 11–15, 2021, Virtual

*Both authors contributed equally to this research. Dongsheng Duan is the corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SIGIR '21, July 11–15, 2021, Virtual Event, Canada

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8037-9/21/07...\$15.00

<https://doi.org/10.1145/3404835.3463060>

Event, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3404835.3463060>

1 INTRODUCTION

Due to serious information overload problem in online services like social networking, news agency and E-commerce, the recommendation has become an important tool to reduce the efforts of online users to find information that they are really interested in. Better recommendation can help attract more user clicks or views, which is one of the most important indicators of online businesses' success. Traditional recommendation models try to infer users' inherent interests based on all their historical behaviors. However, a user's interests might change over time and she might have an intent significantly different from her long-time interests. Moreover, in a temporary session, all we can observe might be only the most recent behaviors for an anonymous user. Therefore, sequential recommendation (SR) has become an import research issue, which tries to recommend the next item that a user would probably be interested in only based on that user's most recent behavior sequence.

The past few years have witnessed plenty of works on SR [9, 14, 16, 19, 20, 25, 27, 33]. However, most existing SR models mainly take the item sequence into consideration and put much emphasis on modeling item-item transition patterns hidden in the sequence, but pay little attention to item attributes. It has been validated that integrating item attributes can improve the performance of traditional recommendation methods, such as collaborative filtering [29], such that attributes are also utilized to improve SR [10, 34, 35]. However, most existing attribute integrated SR models do not directly model the complex relations between items and categorical attributes, as well do not exploit the power of attribute sequence in predicting the next item. It is worth noting that recent work S³-rec [35] considers attribute prediction task in pre-trained model for SR, but it does not directly use attribute sequence to predict the next item.

To overcome the above mentioned limitations of existing models, we propose an **Item Categorical Attribute Integrated Sequential Recommendation (ICAI-SR)** framework, which consists of an Item-Attribute Aggregation (IAA) model and Entity Sequential (ES) models. IAA model explicitly represents the complex relations between items and attributes by using a heterogeneous graph, which regards items and categorical attributes as different types of nodes and items' ownership of attributes as edges. The neighborhood aggregation idea from Graph Neural Network (GNN [1]) is borrowed to model the correlations between items and attributes, and the attention mechanism is utilized to build the aggregation model. ES models include Item Sequential (IS) model and Attribute Sequential (AS) models, and all types of sequential models try to predict the

next item independently. ICAI-SR is instantiated by taking Gated Recurrent Unit (GRU [3]) and Bidirectional Encoder Representation from Transformer (BERT [4]) as ES models, resulting in ICAI-GRU and ICAI-BERT. The main contributions are summarized as follows.

- We propose a general attribute integrated SR framework ICAI-SR, which consists of an Item-Attribute Aggregation (IAA) model and Entity Sequential (ES) Models, such that it can be easily instantiated by different types of IAA and ES models.
- We model the complex relations between items and attributes by carefully designing neighborhood aggregation based on attention mechanism, and utilize attribute sequence in addition to item sequence to predict the next item during model training.
- We instantiate ICAI-SR by taking GRU and BERT as ES models and validate its performance on three public datasets. Results show that it consistently performs better than the basic and competitive models.

2 ICAI-SR FRAMEWORK

The proposed ICAI-SR framework is illustrated in Figure 1. IAA is to compute the item and attribute embeddings from the complex relations between items and attributes, while ES is to discover the entity transition patterns from entity sequences, i.e., item and attribute sequences. IAA and ES can be seamlessly merged together by sharing item and attribute embeddings. In essential, IAA and ES exploit different types of relations between items and attributes, i.e., IAA models the items' ownership of attributes, while ES models item or attribute transition relation in the sequence. For ES models, we build independent sequential models for different types of entities, resulting in one Item Sequential (IS) model and one or more Attribute Sequential (AS) models. Since an item might have multiple attributes of each type, the embeddings of multiple attributes are first aggregated before feeding them into AS models. Besides IS model, AS models are also used to predict the next item during model training. Compared with IS only based SR model, ICAI-SR is more expressive since it not only exploits the complex relations between items and attributes but also performs item prediction based on AS models in addition to IS model.

2.1 Item-Attribute Aggregation Model

We borrow the idea of neighborhood aggregation from GNN to build IAA model. The complex relations between items and attributes form a heterogeneous graph in which an item connects to one or more types of attributes.

IAA is composed of attribute to item aggregation layer and inner attribute aggregation layer illustrated in Figure 2. Before diving into the details of each layer, we first introduce the involved notations. Suppose there are item set I and T attribute types $A = \{A_j | 1 \leq j \leq T\}$ in the item-attribute heterogeneous graph. Following neighborhood sampling idea from one of the most popular GNNs, GraphSAGE [6], we sample a fixed number S of neighboring attributes of each type. The embedding of the i -th item is denoted as $E_i^{(I)}$. The item i 's neighboring attributes of type A_j is denoted by $N_i^{(A_j)}$. The aggregation embeddings of attributes in $N_i^{(A_j)}$ is

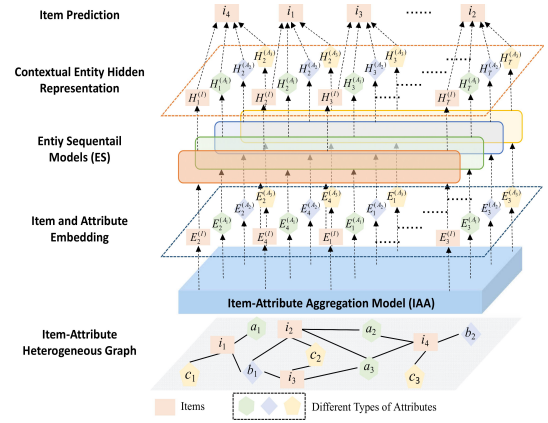


Figure 1: Illustration of ICAI-SR framework, which consists of Item-Attribute Aggregation (IAA) model and Entity Sequential (ES) models.

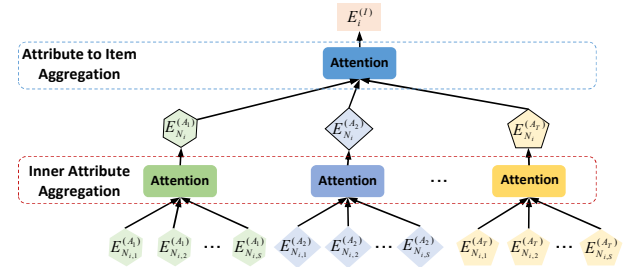


Figure 2: Illustration of IAA, which is composed of attribute to item aggregation and inner attribute aggregation.

denoted by $E_{N_i}^{(A_j)}$. The embedding of the k -th attribute in $N_i^{(A_j)}$ is denoted by $E_{N_{i,k}}^{(A_j)}$.

Attribute to item aggregation layer is to aggregate the embeddings of different attribute types to update item embeddings. For an item i , suppose the aggregation embeddings of all the attribute types of item i , i.e. $E_{N_i}^{(A_j)}$ ($1 \leq j \leq T$), has been computed in advance. Then, the dot product attention is adopted to perform attribute to item aggregation in which attention weights are proportional to the dot products between the embeddings of each attribute type and the initialized embedding of item i . Attribute to item aggregation can be formally expressed as Equation (1).

$$E_i^{(I)} = \tilde{E}_i^{(I)} + \sum_{j=1}^T \beta_j E_{N_i}^{(A_j)},$$

$$\beta_j = \frac{\exp((\tilde{E}_i^{(I)})^\top \cdot E_{N_i}^{(A_j)})}{\sum_{j'=1}^T \exp((\tilde{E}_i^{(I)})^\top \cdot E_{N_i}^{(A_{j'})})}, \quad (1)$$

where β_j is the attention weight of the j -th attribute type of item i , and $\tilde{E}_i^{(I)}$ is the randomly initialized and trainable embedding of item i .

Inner attribute aggregation layer aggregates the embeddings of the same type of attributes of an item, resulting in the embeddings of each attribute type, i.e. $E_{N_i}^{(A_j)}$ ($1 \leq j \leq T$), which can serve

as the input of the attribute to item aggregation layer. The attention mechanism is also utilized to perform this aggregation but using a different way to compute the attention weights. Inspired by Hierarchical Attention Network (HAN [30]), we introduce an attribute level context vector V for different types of attributes, which can be seen as a high-level representation of the fixed query “what is the important attribute”. Following the idea of HAN, $E_{N_i}^{(A_j)}$ is computed by Equation (2).

$$\begin{aligned} E_{N_i}^{(A_j)} &= \sum_{k=1}^S \alpha_{i,k}^{(A_j)} E_{N_{i,k}}^{(A_j)}, \\ U_{N_{i,k}}^{(A_j)} &= \tanh(W E_{N_{i,k}}^{(A_j)} + b), \\ \alpha_{i,k}^{(A_j)} &= \frac{\exp(V^T U_{N_{i,k}}^{(A_j)})}{\sum_{k'=1}^S \exp(V^T U_{N_{i,k'}}^{(A_j)})}, \end{aligned} \quad (2)$$

where $\alpha_{i,k}^{(A_j)}$ is the attention weight of the k -th neighboring A_j typed attribute of item i , and W, b are weights and bias of a dense layer to transform $E_{N_{i,k}}^{(A_j)}$ to a hidden representation $U_{N_{i,k}}^{(A_j)}$.

2.2 Entity Sequential Model

In this paper, we regard items and different types of categorical attributes as entities. The goal of ES models is to exploit the next item’s dependency on different types of historical entity sequences. Due to their effectiveness in modeling sequential data, Recurrent Neural Network (RNN) and Transformer based model [22] have been applied in the sequential recommendation. In this work, we take GRU [3] and BERT [4] as instances of ES models. Given a sequence of entities, ES model takes the embeddings of entities as input and outputs contextual hidden representations for each position in the sequence. Specifically, item sequential model takes the embeddings of items $E^{(I)}$ computed by IAA as input and outputs contextual hidden representations denoted by $H^{(I)}$ for each position in the item sequence, and attribute sequential model takes the aggregation embeddings of an attribute type, e.g. $E^{(A_j)}$, as input and outputs contextual hidden representations denoted by $H^{(A_j)}$ for each position in the A_j typed attribute sequence. For the computational details of instantiated entity sequential model, please refer to GRU4Rec [9] and BERT4Rec [19].

2.3 Model Training And Predicting

The model is trained by minimizing the losses of predicting the next item based on IS and AS models. For an item to be predicted, we first get contextual hidden representation H from the output layer of ES models, and then a dense layer with a softmax activation is used to transform H to predicted probability distribution Y over items. The superscript is also used to distinguish the notations for different types of entities, e.g. $Y^{(I)}$ and $Y^{(A_j)}$ represent predicted probability distribution over items based on IS model and A_j typed AS model respectively. In this paper, the binary cross-entropy loss is adopted as the next entity prediction loss.

Formally, for an item i to be predicted, the prediction loss based on IS model is computed as Equation (3).

$$\mathcal{L}_{IS} = -\log Y_i^{(I)} - \sum_{k \neq i} \log(1 - Y_k^{(I)}). \quad (3)$$

Dataset	ML100K	Beauty	Sports
# Users	943	22,363	35,598
# Items	1674	12,102	18,358
# Actions	99,973	176,139	260,739
# Attr. Types	3	1	1

Table 1: Statistics of the datasets.

Similarly, the prediction loss based on AS models is computed as Equation (4).

$$\mathcal{L}_{AS}^{(A_j)} = -\log Y_i^{(A_j)} - \sum_{k \neq i} \log(1 - Y_k^{(A_j)}), 1 \leq j \leq T. \quad (4)$$

The total loss is computed by putting the prediction losses based on IS and AS models together by using a balancing parameter λ .

$$\mathcal{L} = \mathcal{L}_{IS} + \lambda \sum_{j=1}^T \mathcal{L}_{AS}^{(A_j)}. \quad (5)$$

The training loss $\mathcal{L}$ can be minimized by using stochastic gradient descent or its variations, such as Adam [11]. When performing prediction, only the predicted probability distribution over items based on IS model is used to rank and recommend items.

3 EXPERIMENTS

3.1 Experimental Settings

Datasets. We use three public datasets in the following experiments, which are MovieLens (**ML100K**) [7, 29], and **Beauty** and **Sports** categories from Amazon review datasets [35]. Statistics of the datasets are listed in Table 1. There are three attribute types in ML100K, i.e., genres, directors and actors of movies. For both Beauty and Sports data, there is only one attribute type, i.e., the fine-grained categories of goods.

Evaluation. The last two items in each item sequence are held out for validation and testing purposes respectively, and the remaining items are used for model training. The performance is evaluated by Recall@10, NDCG@10 and MRR, which have been widely adopted by previous studies. To reduce the time consumption of evaluation, following the strategy in [5], we randomly sample 99 negative items for each ground-truth item. Then, each metric is computed based on the ranking of these items and the average metric score over testing data is reported.

Baselines. ICAI-SR takes GRU [3] and BERT [4] as its instantiated ES models. We denote ICAI-SR built upon GRU as **ICAI-GRU** and ICAI-SR built upon BERT as **ICAI-BERT**. **GRU4Rec** and **BERT4Rec** are selected as our baselines. The competitive attribute integrated SR model **FDSA** [34] is selected as a stronger baseline. Although original FDSA considers both categorical and unstructured attributes of items, we only keep its categorical attribute modeling part to make fair comparisons.

Parameter Settings. For GRU4Rec, BERT4Rec and ICAI-SR, the embedding sizes are all set as 256, the batch sizes are all set as 1024, the max length of item sequence is set as 7 on Beauty and Sports data, as 100 on ML100K data. Adam optimizer [11] with the learning rate 0.001 is used to train the model. The balancing parameter λ in ICAI-SR is set as 20. The number of sampled genres, directors and actors of each movie in ML100K data are set as 2, 1 and 3, and the number of sampled single type of attributes in Beauty and Sports

Dataset	Baseline	Recall@10	NDCG@10	MRR
ML100K	GRU4Rec	0.504	0.286	0.241
	ICAI-GRU	0.536	0.309	0.259
	BERT4Rec	0.608	0.338	0.275
	FDSA	0.602	0.326	0.268
	ICAI-BERT	0.622	0.353	0.289
Beauty	GRU4Rec	0.371	0.227	0.201
	ICAI-GRU	0.426	0.263	0.232
	BERT4Rec	0.387	0.241	0.216
	FDSA	0.446	0.280	0.247
	ICAI-BERT	0.466	0.298	0.261
Sports	GRU4Rec	0.322	0.180	0.158
	ICAI-GRU	0.449	0.270	0.235
	BERT4Rec	0.408	0.239	0.208
	FDSA	0.469	0.284	0.246
	ICAI-BERT	0.485	0.289	0.249

Table 2: Recommendation performances of ICAI-SR (including ICAI-GRU and ICAI-BERT), basic SR models GRU4Rec and BERT4Rec, and competitive attribute integrated SR model FDSA.

Baseline	Recall@10	Improve	NDCG@10	Improve	MRR	Improve
w/o IAA and AS	0.408		0.239		0.208	
w/o AS	0.449	10.05%	0.263	10.04%	0.227	9.13%
ICAI-BERT	0.485	18.87%	0.289	20.92%	0.249	19.71%

Table 3: Ablation Study of ICAI-BERT on Sports Data.

data are set as 4 and 6 respectively. The item mask ratio of BERT is set as 0.15 for BERT4Rec and ICAI-BERT. The hyper-parameters of FDSA are all set following the suggestions of the original paper.

3.2 Experimental Results

The performances of ICAI-SR and baseline models are reported in Table 2. From the comparison results, we can see that ICAI-SR consistently performs better on all the three datasets in terms of all evaluation metrics than corresponding basic models, i.e., ICAI-GRU performs better than GRU4Rec, and ICAI-BERT performs better than BERT4Rec, showing that by integrating item categorical attributes we can obtain superior recommendation performance. We also find that the better the basic sequential model is, the better ICAI-SR performs, showing that ICAI-SR depends heavily on the performance of the basic sequential model that it is built upon. Moreover, ICAI-BERT beats FDSA at a significant level, showing that ICAI-SR can be promising to achieve a new state-of-the-art by properly selecting basic sequential recommendation models. The superiority of ICAI-SR over FDSA also partly lies in the additional attribute sequential model which exploits the attribute transition patterns in predicting the next item.

We perform the ablation study to see the effectiveness of the proposed IAA model and AS models. Due to the space limit, only the results of ICAI-BERT on Sports data are reported in Table 3. It can be seen that both IAA and AS contribute to the improvements of ICAI-BERT. Similar results are obtained for other cases.

Last but not the least, we study the sensitivity of ICAI-SR over balancing parameter λ . Figure 3 shows NDCG@10 scores of ICAI-BERT with different λ . When λ is around 20, the best performance is achieved, that is also why we set λ as 20 in the above experiments.

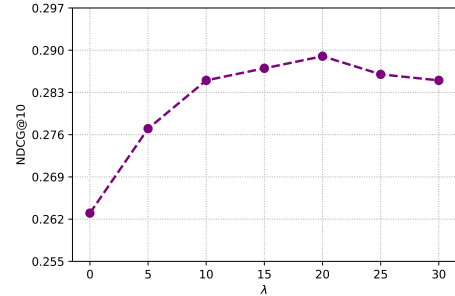


Figure 3: Sensitivity of ICAI-BERT over λ on Sports data.

4 RELATED WORKS

Sequential recommendation can be roughly categorized as Markov chain [8, 18], CNN [20, 28, 31, 32], RNN [9, 26], GNN [16, 25, 27], attention [13, 14, 24, 33], gating [15] and transformer [19] based models. However, these models only take the item sequence into consideration while ignore item attributes. Due to the success of incorporating side information into traditional recommendation models [29], the item attributes have also been integrated in sequential recommendation [10, 17, 21, 34, 35]. The difference of the proposed ICAI-SR from existing attribute integrated SR models is in that we explicitly model the complex relations between items and attributes by carefully designing an item-attribute aggregation model on item-attribute heterogeneous graph, and attribute sequence in addition to item sequence is modeled and utilized to predict the next item during model training.

This work is tightly related to Graph Neural Network (GNN [1]) and attention mechanism [22]. GNN is a powerful tool to model the complex relations between nodes in a graph. Since graph convolution network (GCN [12]) was proposed, GNN has attracted much attention because neural network can be directed built upon the graph structure based on information aggregation [6], which is also extended and widely studied on heterogeneous graphs [2]. In this work, we borrow the idea of information aggregation from GNN to model the complex relations between items and attributes. Due to its effectiveness, the attention mechanism has been applied to a wide spectrum of problem domains, such as language [30], vision and also graph domain [23]. We take advantage of the attention mechanism to build item-attribute aggregation model in this work.

5 CONCLUSION

In this paper, we propose an Item Categorical Attribute Integrated Sequential Recommendation (ICAI-SR) framework, which is composed of an Item-Attribute Aggregation (IAA) model and Entity Sequential (ES) Models. The framework is generic enough so that it can be easily instantiated by different IAA and ES models. By carefully designing attention mechanism based neighborhood aggregation and attribute sequential models, ICAI-SR shows superior recommendation performance over both basic and competitive models.

ACKNOWLEDGMENTS

This work is supported by National Natural Science Foundation of China - Join Fund of Basic Research of General Technology under Grants U1836111 and U1936110.

REFERENCES

- [1] Joan Bruna, Wojciech Zaremba, Arthur Szlam, and Yann Lecun. 2014. Spectral Networks and Locally Connected Networks on Graphs. In *ICLR*.
- [2] Yukuo Cen, Xu Zou, Jianwei Zhang, Hongxia Yang, Jingren Zhou, and Jie Tang. 2019. Representation Learning for Attributed Multiplex Heterogeneous Network. In *SIGKDD*. 1358–1368.
- [3] KyungHyun Cho, Bart van Merriënboer, Dzmitry Bahdanau, and Yoshua Bengio. 2014. On the Properties of Neural Machine Translation: Encoder-Decoder Approaches. *CoRR* abs/1409.1259 (2014).
- [4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL*. 4171–4186.
- [5] Guibing Guo, Shichang Ouyang, Xiaodong He, Fajie Yuan, and Xiaohua Liu. 2019. Dynamic Item Block and Prediction Enhancing Block for Sequential Recommendation. In *IJCAI*. 1373–1379.
- [6] William L Hamilton, Rex Ying, and Jure Leskovec. 2017. Inductive representation learning on large graphs. In *NIPS*.
- [7] F Maxwell Harper and Joseph A Konstan. 2015. The movielens datasets: History and context. *TIIS* (2015), 1–19.
- [8] Ruining He and Julian McAuley. 2016. Fusing similarity models with markov chains for sparse sequential recommendation. In *ICDM*. 191–200.
- [9] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2015. Session-based recommendations with recurrent neural networks. In *ICLR*.
- [10] Balázs Hidasi, Massimo Quadrana, Alexandros Karatzoglou, and Domonkos Tikk. 2016. Parallel recurrent neural network architectures for feature-rich session-based recommendations. In *RecSys*. 241–248.
- [11] Diederik P Kingma and Jimmy Ba. 2015. A method for stochastic optimization. In *ICLR*.
- [12] Thomas N Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *ICLR*.
- [13] Jing Li, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Tao Lian, and Jun Ma. 2017. Neural attentive session-based recommendation. In *CIKM*. 1419–1428.
- [14] Qiao Liu, Yifu Zeng, Refuoc Mokhosi, and Haibin Zhang. 2018. STAMP: short-term attention/memory priority model for session-based recommendation. In *SIGKDD*. 1831–1839.
- [15] Chen Ma, Peng Kang, and Xue Liu. 2019. Hierarchical Gating Networks for Sequential Recommendation. In *SIGKDD*. 825–833.
- [16] Chen Ma, Liheng Ma, Yingxue Zhang, Jianing Sun, Xue Liu, and Mark Coates. 2020. Memory Augmented Graph Neural Networks for Sequential Recommendation. In *AAAI*. 5045–5052.
- [17] Wenjing Meng, Deqing Yang, and Yanghua Xiao. 2020. Incorporating user micro-behaviors and item knowledge into multi-task learning for session-based recommendation. In *SIGIR*. 1091–1100.
- [18] Steffen Rendle, Christoph Freudenthaler, and Lars Schmidt-Thieme. 2010. Factorizing personalized markov chains for next-basket recommendation. In *WWW*. 811–820.
- [19] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *CIKM*. 1441–1450.
- [20] Jiayi Tang and Ke Wang. 2018. Personalized top-n sequential recommendation via convolutional sequence embedding. In *WSDM*. 565–573.
- [21] Trinh Xuan Tuan and Tu Minh Phuong. 2017. 3D convolutional networks for session-based recommendation with content features. In *RecSys*. 138–146.
- [22] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, undefinedukasz Kaiser, and Illia Polosukhin. 2017. Attention is All You Need. In *NuerallIPS*. 6000–6010.
- [23] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, and Y. Bengio. 2017. Graph Attention Networks. In *ICLR*.
- [24] Shoujin Wang, Liang Hu, Longbing Cao, Xiaoshui Huang, Defu Lian, and Wei Liu. 2018. Attention-based transactional context embedding for next-item recommendation. In *AAAI*.
- [25] Ziyang Wang, Wei Wei, Gao Cong, Xiaoli Li, Xianling Mao, and Minghui Qiu. 2020. Global Context Enhanced Graph Neural Networks for Session-based Recommendation. In *SIGIR*.
- [26] Chao-Yuan Wu, Amr Ahmed, Alex Beutel, Alexander J Smola, and How Jing. 2017. Recurrent recommender networks. In *WSDM*. 495–503.
- [27] Shu Wu, Yuyuan Tang, Yanqiao Zhu, Liang Wang, Xing Xie, and Tieniu Tan. 2019. Session-based recommendation with graph neural networks. In *AAAI*. 346–353.
- [28] Xin Xia, Hongzhi Yin, Junliang Yu, Qinyong Wang, Lizhen Cui, and Xiangliang Zhang. 2021. Self-Supervised Hypergraph Convolutional Networks for Session-based Recommendation. *AAAI*.
- [29] Xin Xin, Xiangnan He, Yongfeng Zhang, Yongdong Zhang, and Joemon Jose. 2019. Relational collaborative filtering: Modeling multiple item relations for recommendation. In *SIGIR*. 125–134.
- [30] Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. 2016. Hierarchical Attention Networks for Document Classification. In *NAACL*. 1480–1489.
- [31] Jiaxuan You, Yichen Wang, Aditya Pal, Pong Eksombatchai, Chuck Rosenberg, and Jure Leskovec. 2019. Hierarchical temporal convolutional networks for dynamic recommender systems. In *WWW*. 2236–2246.
- [32] Fajie Yuan, Alexandros Karatzoglou, Ioannis Arapakis, Joemon M Jose, and Xiangnan He. 2019. A simple convolutional generative network for next item recommendation. In *WSDM*. 582–590.
- [33] Shuai Zhang, Yi Tay, Lina Yao, and Aixin Sun. 2018. Next item recommendation with self-attention. *arXiv* (2018).
- [34] Tingting Zhang, Pengpeng Zhao, Yanchi Liu, Victor S Sheng, Jiajie Xu, Deqing Wang, Guanfeng Liu, and Xiaofang Zhou. 2019. Feature-level Deeper Self-Attention Network for Sequential Recommendation.. In *IJCAI*. 4320–4326.
- [35] Kun Zhou, Hui Wang, Wayne Xin Zhao, Yutao Zhu, Sirui Wang, Fuzheng Zhang, Zhongyuan Wang, and Ji-Rong Wen. 2020. S³-rec: Self-supervised learning for sequential recommendation with mutual information maximization. In *CIKM*. 1893–1902.